ELEG 3143 Assignment # 7

1. The pdf of the RV X is given as follows. Find $\mathbb{E}[X]$ and $\mathrm{Var}[X]$.

$$f_X(x) = \begin{cases} \frac{2}{x^2}, & 1 \le x \le 2 \\ 0, & \text{otherwise} \end{cases}$$

2. A random variable has a CDF given by

$$F_X(x) = \begin{cases} A\left[1 - e^{-(x-1)}\right], & x > 1 \\ 0, & x \le 0 \end{cases}$$

   (a) Find the mean $\mathbb{E}[X]$.

   (b) Find the variance $\mathrm{Var}[X]$.

3. A random variable has a pdf given by

$$f_X(x) = \begin{cases} Ax, & 0 < x \le 6 \\ 0, & x \le 0 \end{cases}$$

   (a) Find the value of A.

   (b) Find $\mathbb{E}[X]$.

   (c) Find $\mathrm{Var}[X]$.

   (d) Find the third central moment of X.

   (e) Find the n-th moment, $\mathbb{E}[X^n]$.

4. A random variable has a pdf given by

$$f_X(x) = \begin{cases} ax^2, & 0 < x \le 2 \\ ax, & 2 < x \le 3 \\ 0, & \text{otherwise} \end{cases}$$

(a) Find the value of a.

(b) Find $\mathbb{E}[X]$.

(c) Find $\Pr(2 < X \leq 3)$.

5. Assume a voltage follows Gaussian distribution with mean 5 and variance 16, that is $V \sim \mathcal{N}(5, 16)$.

   (a) What is the probability that a measured voltage is greater than 0?

   (b) What is the probability that a measured voltage is greater than 0 and less than or equal to the mean value?

   (c) What is the probability that a measured voltage is greater than twice of the mean value?

6. Prove the mean of a Gaussian random variable with the following pdf is m.

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right) \tag{1}$$